

2207

نموذج إجابة

جمهورية مصر العربية
وزارة التربية والتعليم والتعليم الفني
امتحان دبلوم مدراس التكنولوجيا التطبيقية الصناعية
(نظام الثلاث سنوات)

دور أول - عام 2023

الزمن: ساعتان

المادة: Physics

التخصص : جميع التخصصات الصناعية لمدراس (WE – HST)

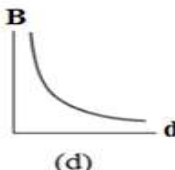
Answer the following groupsGroup 1 :

Write the scientific term expressed by the following statements:

	Statement
1	Tesla
2	Eddy currents
3	Self-induction of a coil
4	Efficiency of a transformer
5	Black body radiation
6	Surface potential barrier
7	Work function
8	X- rays
9	Stimulated Emission
10	Logic gates

Group 2 :

Put circle around the correct answer

11	(b) doubled
12	
13	(a) increase
14	(b) 1 m
15	(c) 200V
16	(a) 300 K
17	(d) the width of depletion layer increases and the diode resistance increases
18	(c) they can be deflected under the effect of electric and magnetic fields
19	(b) the energy of the incident photon is greater than the work function of the metal
20	(d) the output-to-input electric power ratio is smaller than one

Group 3 :

Correct the underlined in the following statements

	Statement
21	Perpendicular to Normal to
22	Increasing
23	I A N
24	Henery Weber/Ampere Ohm. second Volt.second/Ampere
25	Greater than
26	decreases
27	Subject (obey)
28	Increase

Group 4 :

Answer the following

	problem	
29 30	When the current changes from 5A to 2A in 0.05 second, an e m f of 15 V is induced in a coil. Calculate the coefficient of self-induction in the coil.	$\text{emf} = -L \times \frac{\Delta I}{\Delta t}$ $15 = -L \times \frac{(2 - 5)}{0.05}$ $L = 0.25 \text{ H}$
31 32	A rectangular coil of 500 turns has dimensions 0.2m × 0.1 m, carries a current 5A. The plane of the coil makes angle 60° to a uniform magnetic field of flux density 0.8 T. Calculate the magnetic torque acting on the coil	$\tau = B I A N \sin \theta$ $\tau = 0.8 \times 5 \times (0.2 \times 0.1) \times 500 \times \sin (30)$ $\tau = 20 \text{ N.m.}$
33 34	A metallic ring is placed in a uniform magnetic field perpendicular to the plane of the ring. If the magnetic flux changes from 0.05 Wb to - 0.05 in 0.05 s. Calculate the average induced EMF in the loop	$\text{emf} = \frac{-N \Delta \Phi_m}{\Delta t}$ $\text{emf} = \frac{(-1)(-0.05 - 0.05)}{0.05}$ $\text{emf} = 2 \text{ V}$
35 36	A metal has a work function $3.85 \times 10^{-19} \text{ J}$. Calculate the maximum kinetic energy of electrons emitted from the metal when light of frequency $7.5 \times 10^{14} \text{ Hz}$ is shone on the surface (Given that $h = 6.6 \times 10^{-34} \text{ J.s}$)	$K.E = h\nu - E_w$ $K.E = (6.6 \times 10^{-34} \times 7.5 \times 10^{14}) - (3.85 \times 10^{-19})$ $K.E = 1.1 \times 10^{-19} \text{ J}$
37 38	The current gain (β) of a transistor in common emitter mode is 50. Calculate the collector current if the base current $50 \mu\text{A}$	$\beta = \frac{I_C}{I_B}$ $50 = \frac{I_C}{50}$ $I_C = 2500 \mu\text{A} = 2.5 \times 10^{-3} \text{ A}$

Group 5:**Mention one factor only that affects each of the following:**

	Statement
39	a) intensity of current passing through the solenoid b) number of turns c) magnetic permeability d) the length of the solenoid
40	a) magnetic flux density b) the current intensity c) the area of the coil's plane d) number of turns of the coil
41	a) The presence of an iron core inside the coil. b) The volume of the coil and the number of its turns. c) The distance separating them.
42	a) the length of wire (l) b) density of the magnetic field (B) c) velocity by which the wire move (v)
43	the potential difference between the cathode and anode
44	the potential difference between the cathode and anode

Group 6:**Compare between each of the following**

45 46	Point of comparison	NOR gate			NAND gate		
	Complete the truth table	Input A	Input B	Output	Input A	Input B	Output
		1	1	0	1	1	0
47 48	Point of comparison	Direct current (DC).			Alternating current (AC)		
	The direction	One direction			Two opposite directions		
49 50	Point of comparison	Lenz's rule			The right Cork Screw rule		
	The use	used to determine the direction of induced current in a coil			used to determine the direction of the magnetic field resulting from an electric current in a circular coil		